

Lab Assignment 6

PCS221 Cloud Computing – Map Reduce (Without EMR)

Instructions:

For each question, implement using:

(A) Manual Python approach (Mapper → Shuffle → Reducer)

(B) mrjob framework

Submit .ipynb file on Github

Q1.	Word Count - Count the frequency of each word Input: • <i>hadoop is fast</i> • <i>hadoop is scalable</i>
Q2.	Character Count - Count the frequency of each character (ignore spaces). Input: <i>big data</i>
Q3.	Average Word Length (Per Word) - Compute the average length of each word. Input: <i>data science data big data</i>
Q4.	Global Average Word Length - Compute the average length of all words. Input: <i>hadoop mapreduce spark</i>
Q5.	Perform Q1-Q4 on the file available on the link: https://drive.google.com/file/d/16TlgKhcc2JH8jyJXfwouOV3y7ACX_aas/view?usp=sharing • Also find Top 5 most frequent words
Q6.	Compute average marks for each student. Input: <i>A 80</i> <i>B 70</i> <i>A 90</i> <i>B 60</i> <i>A 100</i>
Q7.	Compute average salary per department and Highest Paid Department (Based on Average Salary) Input: <i>HR 50000</i> <i>IT 70000</i> <i>HR 60000</i> <i>IT 80000</i>
Q8.	Computer average temperature per country <i>New York,38</i> <i>London,29</i> <i>Tokyo,35</i> <i>New York,32</i> <i>Delhi,45</i> <i>Ambala,35</i>
Q9.	Redo Q8 with the dataset available on: https://www.kaggle.com/datasets/heemalichaudhari/global-land-temperatures